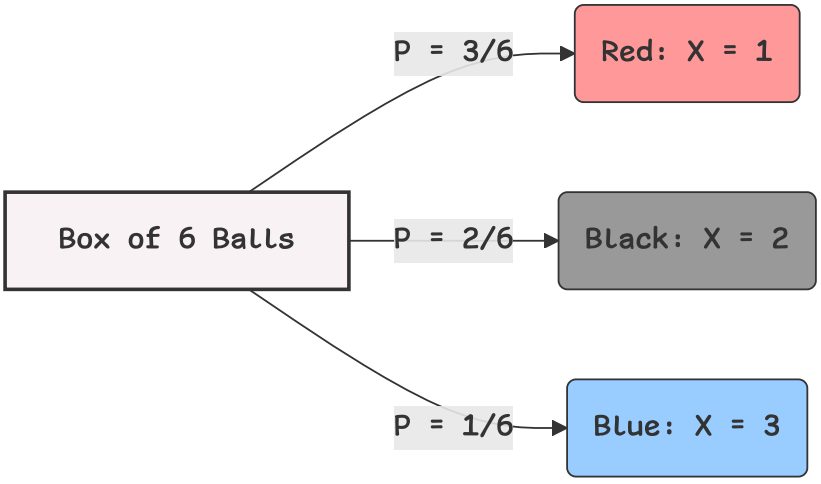
2 - Review

Probability and Random Variables Review

1. Fundamentals of Probability

- **Probability:** The chance of an event occurring in an experiment.
- **Experiment Example:** Choosing a ball from a box.
  - If a box has 3 red, 1 blue, and 2 black balls, the probabilities are: $P(\text{Red}) = \frac{3}{6}$, $P(\text{blue}) = \frac{1}{6}$, $P(\text{black}) = \frac{2}{6}$.
- **Random Variable (R.V.):** A function that assigns numbers to events.
  - Using the previous example, we can assign $X = 1$ for Red, $X = 2$ for Black, and $X = 3$ for Blue.



2. Discrete Random Variables

A discrete R.V. takes on distinct, countable values.

- Probability Mass Function (PMF): Defines the probability of the variable equaling a specific value.

$$P_X(x) = \text{Prob}(X = x)$$

- Cumulative Distribution Function (CDF): Defines the probability that the variable is less than or equal to a specific value.

$$P_X(x) = \text{Prob}(X \leq x)$$

- Expectation (Average/Mean):

$$E(X) = \sum_i x_i \cdot P_X(x_i)$$

- Variance: The average of the squares minus the square of the average.

$$\text{Var}(X) = \sigma_x^2 = E(X^2) - (E(X))^2$$

  - Where $E(X^2) = \sum_i x_i^2 \cdot P_X(x_i)$.
- Standard Deviation:

$$\text{Std}(X) = \sigma_x = \sqrt{\text{Var}} = \sqrt{\sigma_x^2}$$

3. Continuous Random Variables

A continuous R.V. takes on continuous values.

- Probability Density Function (PDF): $P_X(x)$.
- Cumulative Distribution Function (CDF):

$$P_X(x) = \text{Prob}(X \leq x) = \int_{-\infty}^{x} P_X(x)dx$$

  - Useful CDF Properties:
    - $P(X > x) = 1 - P(X \leq x) = 1 - P_X(x)$
    - $P(a \leq X \leq b) = P_X(b) - P_X(a) = \int_a^b P_X(x)dx$
- Expectation:

$$E(X) = \int_{-\infty}^{\infty} x \cdot P_X(x)dx$$

- Variance:

$$\text{Var}(X) = \sigma_x^2 = E(X^2) - (E(X))^2 = \int_{-\infty}^{\infty} x^2 P_X(x)dx - \left(\int_{-\infty}^{\infty} xP_X(x)dx\right)^2$$

4. Important Probability Rules

- Product Rule: $P(A|B) = \frac{P(A,B)}{P(B)} \Rightarrow P(A,B) = P(A|B) \cdot P(B)$
- Bayes' Rule: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- Conditional Expectation: $E[X] = E[X|A]P(A) + E[X|B]P(B)$

5. Exponential Random Variable

We are interested in the Exponential R.V. because it is commonly used to model inter-arrival times.

- PDF: $f_X(x) = \mu e^{-\mu x}$ if $x \geq 0$, and 0 if $x \leq 0$.
- CDF:

$$F_X(x) = P\{X \leq x\} = \begin{cases} 1 - e^{-\mu x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

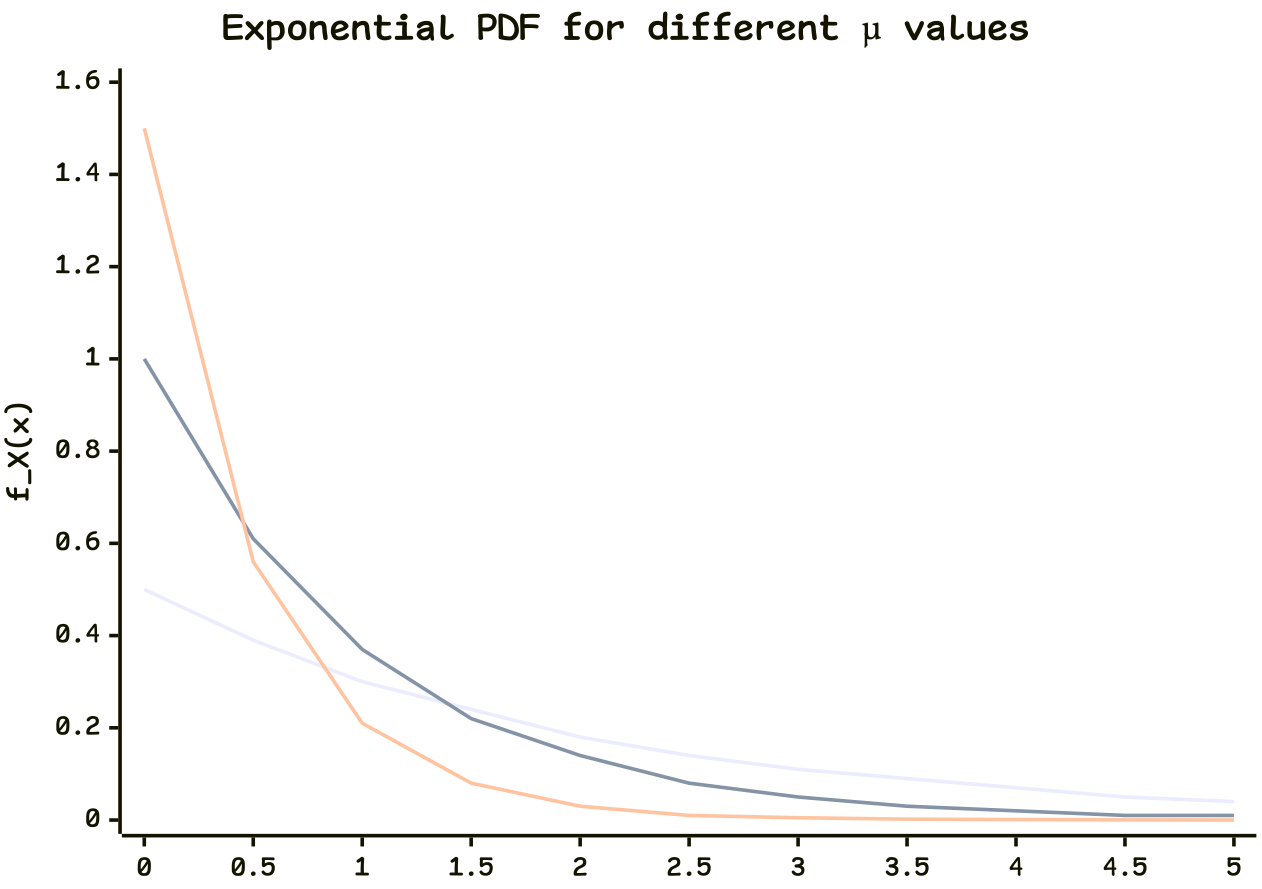
- Mean & Variance:
  - $E[X] = \frac{1}{\mu}$
  - $\text{Var}(X) = \frac{1}{\mu^2}$
- Memoryless Property: The distribution does not "remember" past time spent.

$$P(X > x + t | X > t) = P(X > x)$$

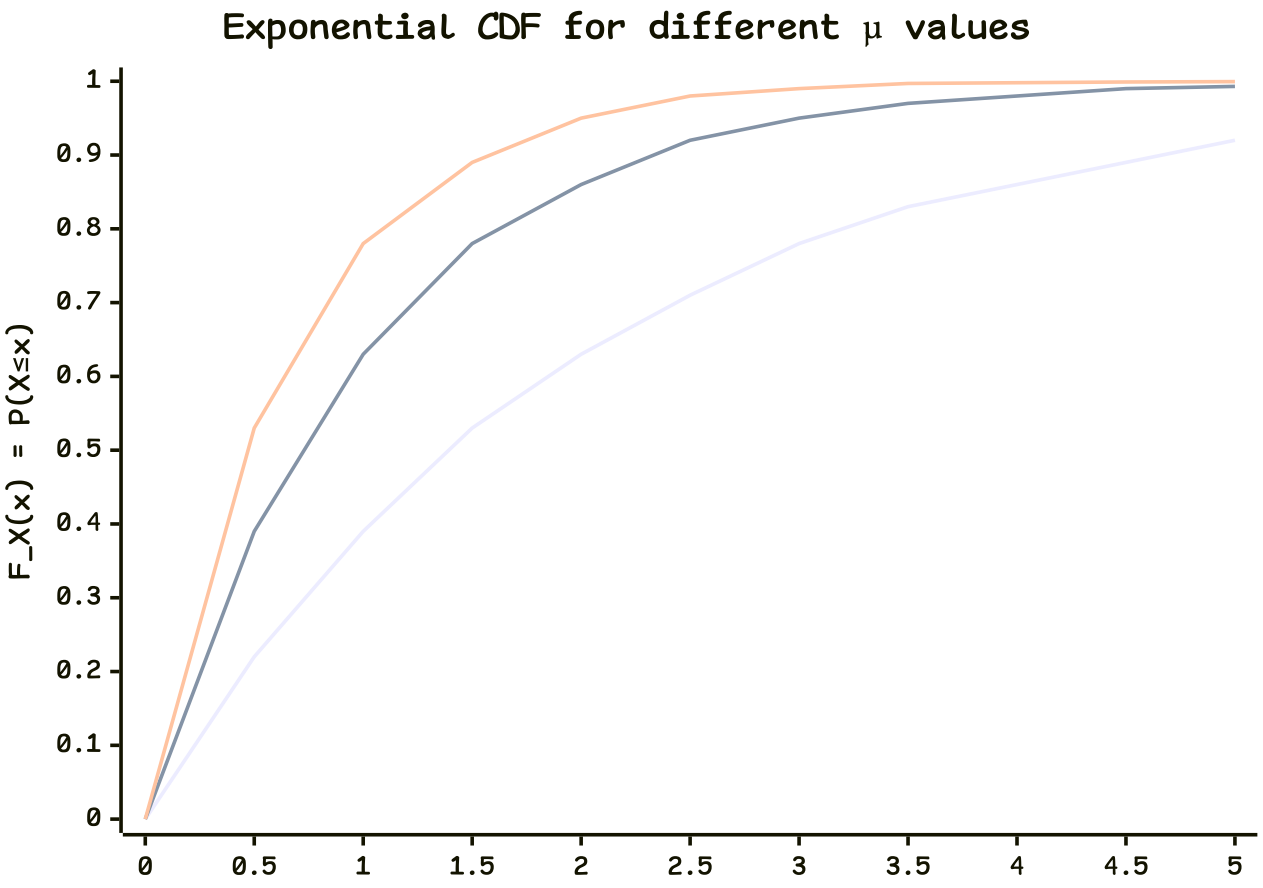
  - *Proof outline:* $\frac{P(X>x+t)}{P(X>t)} = \frac{e^{-\mu(x+t)}}{e^{-\mu t}} = e^{-\mu x}$.

Exponential Distribution Graphs

PDF: $f_X(x) = \mu e^{-\mu x}$



CDF: $F_X(x) = 1 - e^{-\mu x}$


Exponential CDF for different μ values
F_X(x) = P(X≤x)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5

6. Poisson Random Variable

A discrete R.V. that defines the probability of a random event occurring k times over an interval. It is well known as a good model for arrivals in networks.

- PMF:

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

  where $k = 0, 1, 2, 3 \ldots$
- Mean & Variance:
  - λ represents the average (or expected) number of events over the given interval.
  - $E[X] = \lambda$
  - $\mathrm{Var}(X) = \lambda$
- Relationship to Exponential R.V.: For a Poisson R.V. X, the inter-arrival time between events is an Exponential R.V..
  - $P(T \leq s) = 1 - P(X = 0) = 1 - e^{-\lambda s}$, which matches the CDF of an Exponential R.V..

7. Solved Examples

Example 1: Expectation and Variance

Suppose X has the following PMF: $p(0) = 0.2$, $p(1) = 0.5$, $p(2) = 0.3$.

Answer:

- $E[X] = 0(0.2) + 1(0.5) + 2(0.3) = 1.1$
- Let $Y = X^2$. $E[X^2] = E[Y] = 0(0.2) + 1(0.5) + 4(0.3) = 1.7$
- $\mathrm{Var}[X] = E[X^2] - (E[X])^2 = 1.7 - (1.1)^2 = 0.49$

Example 2: Exponential Memoryless Property

Let X be the amount of time a customer spends in a bank. X is exponentially distributed with a mean of 10 minutes.

- Q1: What is the probability she spends more than 15 minutes?

  Answer:
  X is exp (memoryless) -> $E[X] = 10 \Rightarrow \mu = \frac{1}{10}$

  $P(X > 15) = e^{-15/10} = e^{-3/2}$
- Q2: Given that the customer is still in the bank after 10 minutes, what is the probability she will spend more than 15 minutes total?

  Answer: Because of the memoryless property, this is equivalent to finding the probability of spending at least 5 *more* minutes:
  $P(X > 15 - 10) = P(X > 5) = e^{-5/10} = e^{-1/2}$.

Example 3: Poisson Distribution

If the number of accidents occurring on a highway each day is a Poisson random variable with parameter $\lambda = 3$. What is the probability that no/1 accidents occur today?

Given:

- $\lambda = 3$ (average number of accidents per day)
- $E[X] = \lambda = 3$
- $\mathrm{Var}(X) = \lambda = 3$

PMF Formula:

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

Answer:

Q1: Probability of 0 accidents today:

$$P(X = 0) = e^{-\lambda} \frac{\lambda^0}{0!} = e^{-3} \frac{3^0}{1} = e^{-3} \cdot 1 = e^{-3}$$

$$P(X = 0) = e^{-3} \approx 0.0498 \approx 4.98\%$$

Q2: Probability of 1 accident today:

$$P(X = 1) = e^{-\lambda} \frac{\lambda^1}{1!} = e^{-3} \frac{3^1}{1} = e^{-3} \cdot 3 = 3e^{-3}$$

$$P(X = 1) = 3e^{-3} \approx 0.1494 \approx 14.94\%$$

Example 4: Poisson and Exponential

People enter a restaurant at a Poisson rate $\lambda = 1$ per day.

- Q: What is the probability that the elapsed time between the 10th and 11th arrival exceeds two days?

Answer:

The inter-arrival time between the 10th and 11th arrival (τ_{10}) is an exponential random variable with rate $\mu = \lambda = 1$.

Using the relationship $P(X > x) = 1 - F_X(x) = 1 - (1 - e^{-\mu x}) = e^{-\mu x}$:

$$P\{\tau_{10} > 2\} = 1 - P\{\tau_{10} \leq 2\} = 1 - F_\tau(2)$$

$$= 1 - (1 - e^{-\mu \cdot 2}) = 1 - (1 - e^{-1 \times 2})$$

$$= 1 - (1 - e^{-2}) = e^{-2}$$

$$\boxed{P(T_{10} > 2) = e^{-2} \approx 0.1353 \approx 13.53\%}$$

Example 5: Discrete R.V. with PMF and CDF

You are monitoring the number of customers at a bank at 9, 10, 11 AM for four days:

Days	9 AM	10 AM	11 AM
1 (Sun)	10	8	16
2 (Mon)	8	8	10
3 (Tue)	10	16	8
4 (Wed)	10	10	10

Define the R.V.: Let X = number of customers at the bank. X takes values $\{8, 10, 16\}$.

PMF:

Answer:

- $P(X = 8) = \frac{4}{12}$ (occurs 4 times out of 12 observations)
- $P(X = 10) = \frac{6}{12}$ (occurs 6 times out of 12 observations)
- $P(X = 16) = \frac{2}{12}$ (occurs 2 times out of 12 observations)

CDF:

Answer:

$$F_X(x) = P(X \leq x) = \begin{cases} 0 & x < 8 \\ \frac{4}{12} & 8 \leq x < 10 \\ \frac{10}{12} & 10 \leq x < 16 \\ 1 & x \geq 16 \end{cases}$$

Expectation:

Answer:

$$E[X] = \sum_i x_i \cdot P(x_i) = 8 \cdot \frac{4}{12} + 10 \cdot \frac{6}{12} + 16 \cdot \frac{2}{12} = \frac{32 + 60 + 32}{12} = \frac{124}{12} \approx 10.33$$

Variance:

Answer:

$$E[X^2] = 8^2 \cdot \frac{4}{12} + 10^2 \cdot \frac{6}{12} + 16^2 \cdot \frac{2}{12} = \frac{256 + 600 + 512}{12} = \frac{1368}{12} = 114$$

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = 114 - \left(\frac{124}{12}\right)^2 = 114 - \frac{15376}{144} = \frac{16416 - 15376}{144} = \frac{1040}{144} \approx 7.22$$

8. Additional Example: Car Battery Life

Problem: Suppose that the number of miles that a car can run before its battery wears out is exponentially distributed with an average value of 10,000 miles. If a person desires to take a 5,000-mile trip, what is the probability that he or she will be able to complete the trip without having to replace the car battery?

What can be said when the distribution is not exponential?

Answer:

Let X be the battery lifetime (in thousands of miles). X is exponentially distributed with:

$$E[X] = 10 \text{ (thousand miles)}$$

Since $E[X] = \frac{1}{\mu}$ for an exponential distribution:

$$\mu = \frac{1}{10,000} \text{ per mile}$$

We want to find the probability that the battery lasts more than 5,000 miles:

$$P(X > 5) = 1 - P(X \leq 5) = 1 - F_X(5)$$

Using the CDF:

$$P(X > 5) = 1 - (1 - e^{-\mu \cdot 5}) = e^{-\frac{1}{10} \cdot 5} = e^{-\frac{1}{2}}$$

$$P(X > 5) = e^{-0.5} \approx 0.6065 \approx 0.604$$

The probability of completing the 5,000-mile trip without replacing the battery is approximately 60.4% (or exactly $e^{-1/2}$).

Discussion: What if the distribution is NOT exponential?

If the battery lifetime does not follow an exponential distribution:

1. We cannot use the memoryless property – the probability of surviving an additional 5,000 miles depends on how much the battery has already been used.
2. We would need to know the specific distribution (e.g., Weibull, Normal, etc.) to calculate survival probabilities.
3. The mean alone is insufficient – we would need more parameters (variance, shape parameters) to fully characterize the distribution.
4. Real-world batteries typically have a "bathtub curve" failure rate: higher failure at the beginning (infant mortality) and end (wear-out) of life, with lower failure in between. This is better modeled by a Weibull distribution than an exponential distribution.

Appendix: Derivations for Exponential Random Variable

Derivation of Mean $E[X]$

$$E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx = \int_0^{\infty} x \cdot \mu e^{-\mu x}\, dx$$

Using integration by parts: Let $u = x$ and $dv = \mu e^{-\mu x}\, dx$

Then $du = dx$ and $v = -e^{-\mu x}$

$$E[X] = \mu \int_0^{\infty} x e^{-\mu x}\, dx = \mu \left[-\frac{x e^{-\mu x}}{\mu} \Big|_0^{\infty} + \int_0^{\infty} \frac{e^{-\mu x}}{\mu}\, dx \right]$$

$$= \mu \left[-\frac{x e^{-\mu x}}{\mu} \Big|_0^{\infty} + \frac{1}{\mu} \int_0^{\infty} e^{-\mu x}\, dx \right]$$

Evaluating the first term:

- At $x = \infty$: $\lim_{x \to \infty} x e^{-\mu x} = 0$ (exponential decay beats linear growth)
- At $x = 0$: $0 \cdot e^0 = 0$

So the first term equals $0 - 0 = 0$.

For the second term:

$$\int_0^{\infty} e^{-\mu x}\, dx = \left[-\frac{e^{-\mu x}}{\mu} \right]_0^{\infty} = 0 - \left(-\frac{1}{\mu} \right) = \frac{1}{\mu}$$

Therefore:

$$E[X] = \mu \cdot \frac{1}{\mu} \cdot \frac{1}{\mu} = \frac{1}{\mu}$$

$$\boxed{E[X] = \frac{1}{\mu}}$$

Derivation of $E[X^2]$

$$E[X^2] = \int_0^\infty x^2 \cdot \mu e^{-\mu x}\, dx$$

Using integration by parts twice (first with $u = x^2$, $dv = \mu e^{-\mu x} dx$):

$$E[X^2] = \left[-x^2 e^{-\mu x}\right]_0^\infty + 2\int_0^\infty x e^{-\mu x}\, dx$$

The first term evaluates to 0 (at both limits). For the second term, we already computed $\int_0^\infty x e^{-\mu x}\, dx = \frac{1}{\mu^2}$ in the mean derivation.

$$E[X^2] = 0 + 2 \cdot \frac{1}{\mu^2} = \frac{2}{\mu^2}$$

$$\boxed{E[X^2] = \frac{2}{\mu^2}}$$

Derivation of Variance

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = \frac{2}{\mu^2} - \left(\frac{1}{\mu}\right)^2 = \frac{2}{\mu^2} - \frac{1}{\mu^2}$$

$$\boxed{\mathrm{Var}(X) = \frac{1}{\mu^2}}$$

Derivation of the Memoryless Property

The memoryless property states:

$$P\{X > x + t \mid X > t\} = P\{X > x\}$$

Part (A): Right-Hand Side (RHS)

$$P\{X > x\} = 1 - P(X \leq x) = 1 - (1 - e^{-\mu x}) = e^{-\mu x}$$

Part (B): Left-Hand Side (LHS)

Using the definition of conditional probability:

$$P\{X > x + t \mid X > t\} = \frac{P\{X > x + t, X > t\}}{P(X > t)}$$

Since if $X > x + t$ (and $x > 0$), then automatically $X > t$, so:

$$P\{X > x + t, X > t\} = P\{X > x + t\}$$

Therefore:

$$P\{X > x + t \mid X > t\} = \frac{P\{X > x + t\}}{P(X > t)}$$

Now compute each probability:

$$P\{X > x + t\} = 1 - P(X \leq x + t) = 1 - (1 - e^{-\mu(x+t)}) = e^{-\mu(x+t)}$$

$$P\{X > t\} = 1 - P(X \leq t) = 1 - (1 - e^{-\mu t}) = e^{-\mu t}$$

Substituting:

$$P\{X > x + t \mid X > t\} = \frac{e^{-\mu(x+t)}}{e^{-\mu t}} = \frac{e^{-\mu x} \cdot e^{-\mu t}}{e^{-\mu t}} = e^{-\mu x}$$

Conclusion:

$$\text{LHS} = e^{-\mu x} = \text{RHS}$$

Therefore, X is memoryless.

$$\boxed{P(X > x + t \mid X > t) = P(X > x) = e^{-\mu x}}$$

Proof That Inter-Arrival Time is Exponential

This proof demonstrates the relationship between the Poisson and Exponential distributions.

Setup:

- Let X be a Poisson random variable representing the number of arrivals/events in time s
- Let τ_n be the inter-arrival time between event n and event $n+1$
- λ = arrival rate (events per unit time)

Timeline:

```
Event                    No Events                    Event
   |<------------------- τ_n ------------------->|
t_n                                              t_{n+1}
```

Finding the CDF of τ:

The probability that the inter-arrival time is less than or equal to s is:

$$P\{\tau_n \leq s\} = 1 - P(\text{no arrivals in time } s)$$

Since the number of arrivals follows a Poisson distribution:

$$P\{\tau_n \leq s\} = 1 - P\{X = 0\}$$

Using the Poisson PMF with $k = 0$:

$$P\{X = 0\} = e^{-\lambda s}\frac{(\lambda s)^0}{0!} = e^{-\lambda s} \cdot \frac{1}{1} = e^{-\lambda s}$$

Therefore:

$$P\{\tau_n \leq s\} = 1 - e^{-\lambda s}$$

Conclusion:

The CDF $F_\tau(s) = 1 - e^{-\lambda s}$ is exactly the CDF of an Exponential random variable with parameter $\mu = \lambda$.

$$\boxed{\text{The inter-arrival time } \tau \text{ follows an Exponential distribution with rate } \lambda}$$